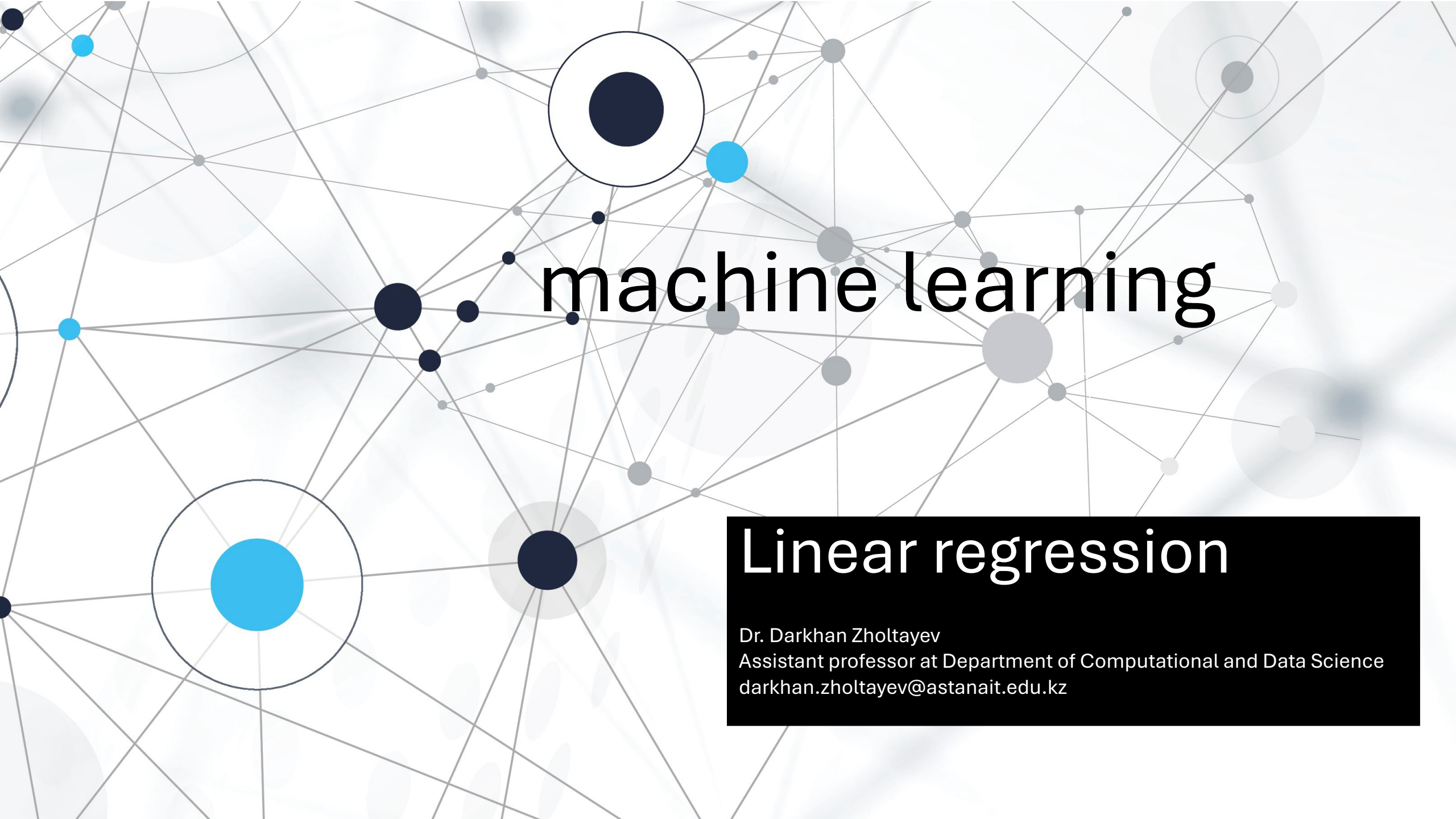


The background of the slide features a complex network diagram. It consists of numerous nodes of varying sizes and colors (dark blue, light blue, and grey) interconnected by a web of thin, light grey lines. Some nodes are highlighted with larger, concentric circles. The overall aesthetic is modern and technological.

machine learning

Linear regression

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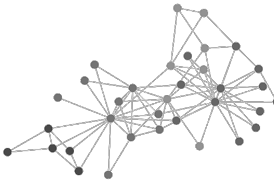
Topics to cover

What is the regression

Linear regression

Least square error

Methods to find parameters



General graph

AI map



What is the regression

Regression is a statistical method used in machine learning and data analysis to model and analyze the relationships between a dependent variable (also known as the output or target) and one or more independent variables (also known as predictors or features). The goal of regression is to predict the value of the dependent variable based on the values of the independent variables.

Some examples are:

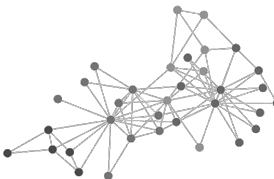
- Linear Regression: Assumes a linear relationship between the dependent and independent variables.
- Polynomial Regression: Models non-linear relationships by transforming the independent variable into higher powers.
- Logistic Regression: Used for binary classification tasks, predicting categorical outcomes instead of continuous values.

In general, regression analysis helps in identifying the strength and nature of relationships between variables and making predictions based on these relationships.



Linear Regression

- Technique used for the modeling and analysis of numerical data
- Exploits the relationship between two or more variables so that we can gain information about one of them through knowing values of the other
- Regression can be used for prediction, estimation, hypothesis testing, and modeling causal relationships



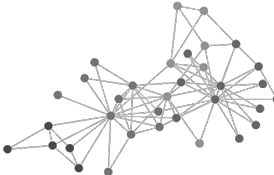
Example Problem

Meddicorp Sales

Meddicorp Company sells medical supplies to hospitals, clinics, and doctor's offices.

Meddicorp's management considers the effectiveness of a new advertising program.

Management wants to know if the **advertisement** in 1999 is related to **sales**.



Data

The company observes for 25 offices the yearly sales (in thousands) and the advertisement expenditure for the new program (in hundreds)

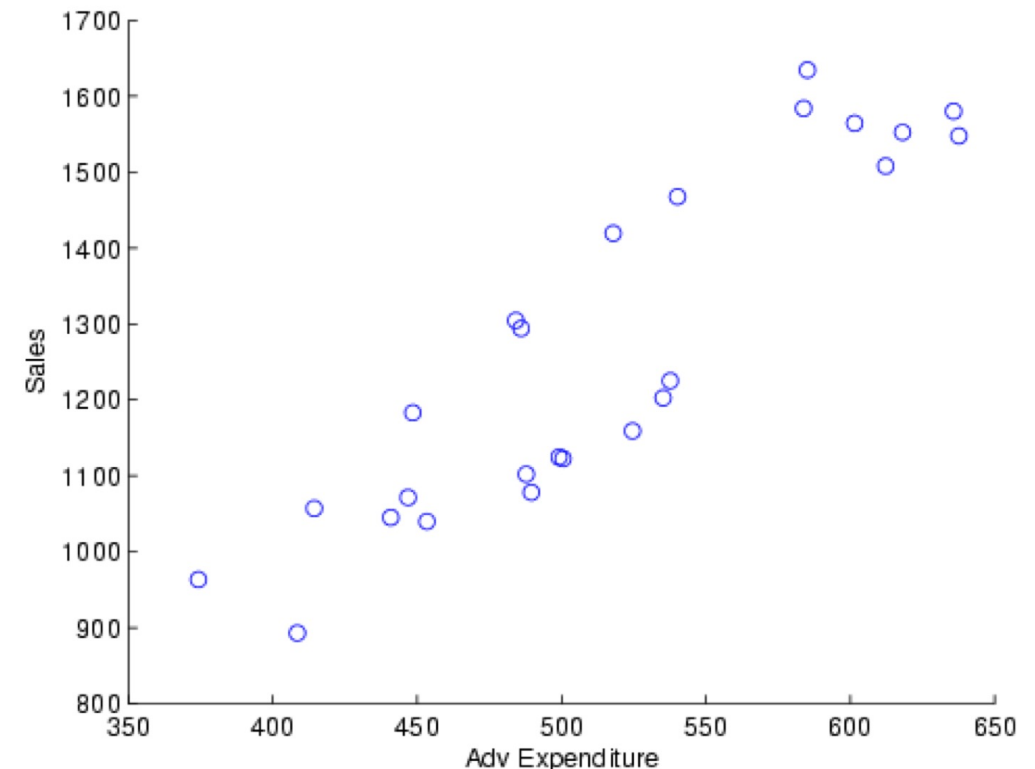
	SALES	ADV
1	963.50	374.27
2	893.00	408.50
3	1057.25	414.31
4	1183.25	448.42
5	1419.50	517.88



Data

The company observes for 25 offices the yearly sales (in thousands) and the advertisement expenditure for the new program (in hundreds)

	SALES	ADV
1	963.50	374.27
2	893.00	408.50
3	1057.25	414.31
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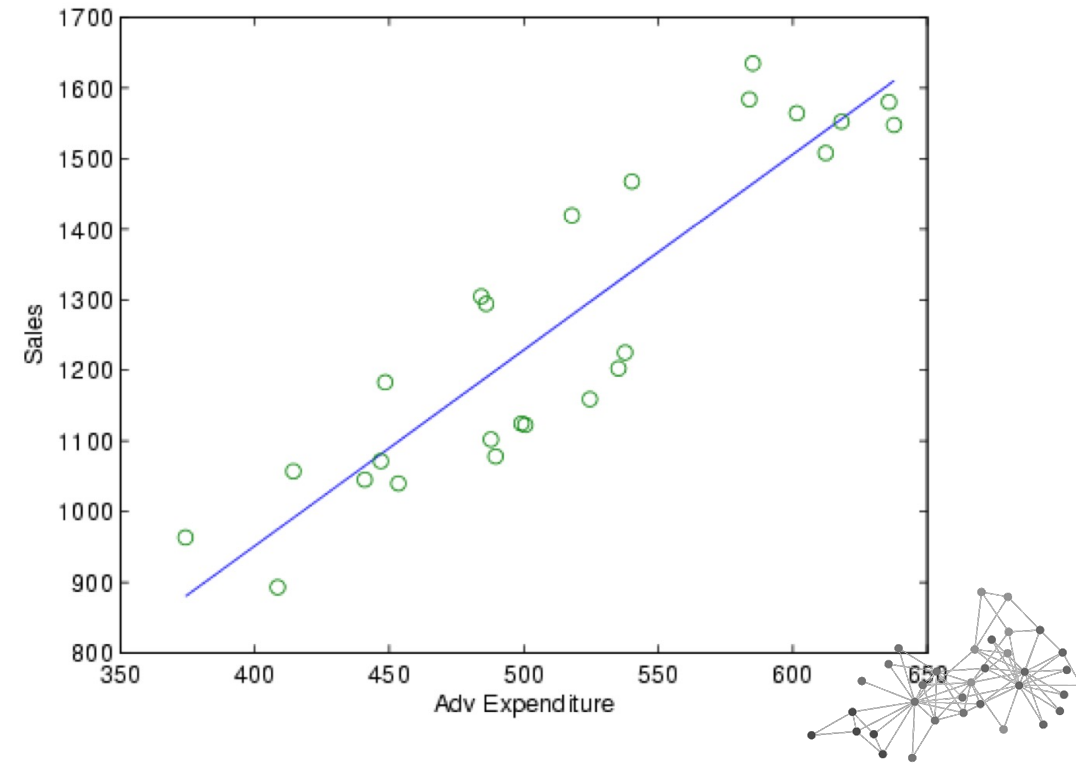
Xie, Y. (2013). *Lecture 11: Simple Linear Regression*. H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology. Retrieved from [<https://www2.isye.gatech.edu/~yxie77/isye2028/lecture11.pdf>].



Data

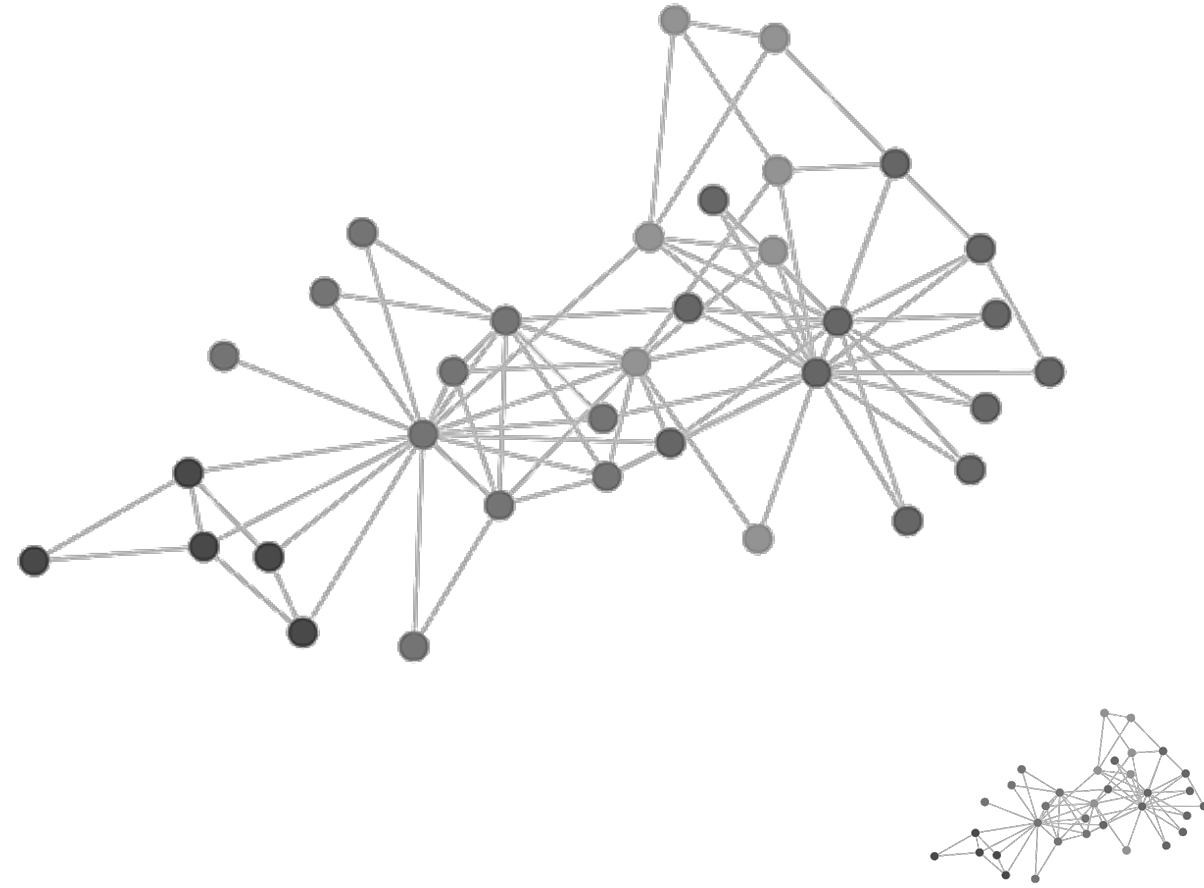
The company observes for 25 offices the yearly sales (in thousands) and the advertisement expenditure for the new program (in hundreds)

	SALES	ADV
1	963.50	374.27
2	893.00	408.50
3	1057.25	414.31
4	1183.25	448.42
5	1419.50	517.88



Linear Regression

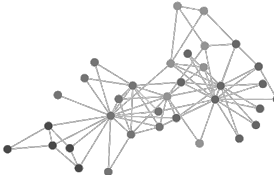
- The collection of statistical tools that are used to model and explore relationships between variables that are related in nondeterministic manner is called regression analysis
- Occurs frequently in engineering and science



Definition: Linear regression

- We observe a **response** or **dependent** variable (Y)
- With each (Y), we also observe **regressors** or **predictors** $\{X_1, \dots, X_n\}$
- Goal: determine the mathematical relationship between response variables and regressors

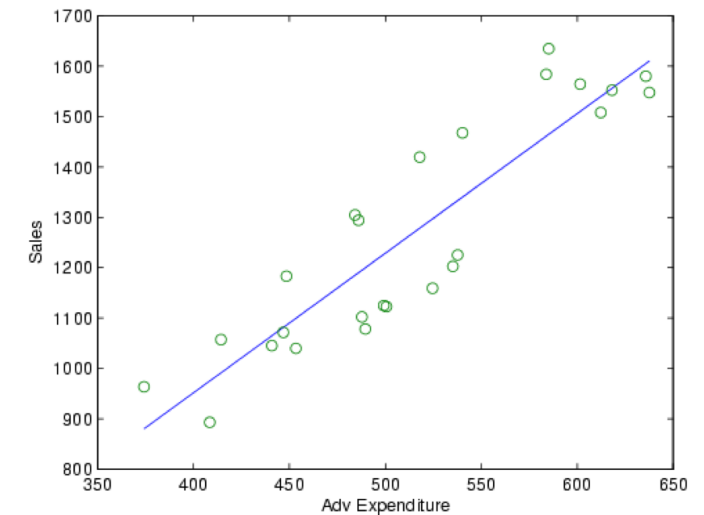
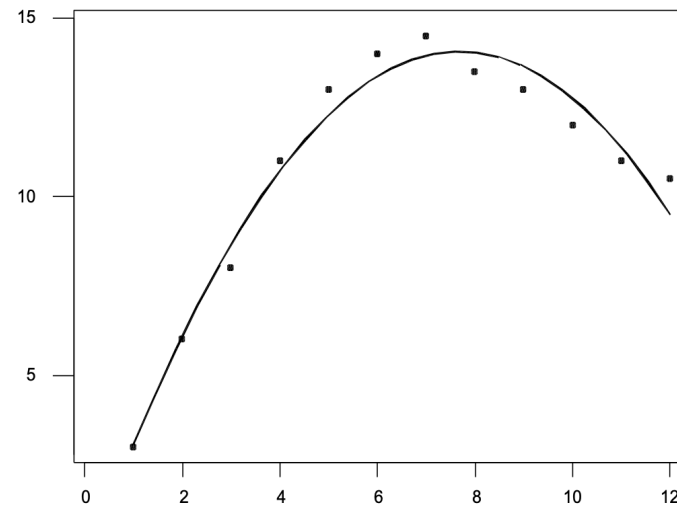
- $$Y = h(X_1, \dots, X_n)$$



Linear regression

- Function can be non-linear
- In this class, we will focus on the case where Y is a linear function of $\{X_1, \dots, X_n\}$

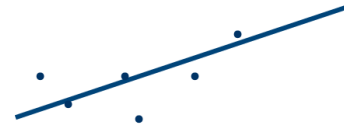
$$Y = h(X_1, \dots, X_n) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n$$



Linear regression: different forms

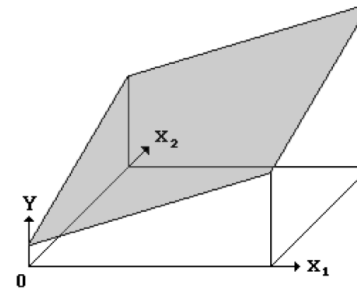
- Simple linear regression

$$Y = \beta_0 + \beta_1 X + \varepsilon$$



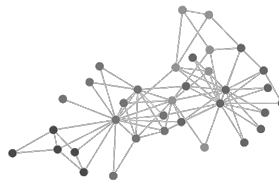
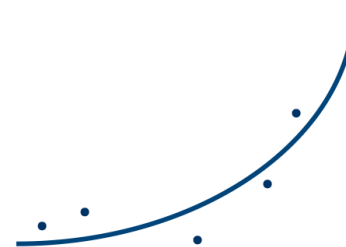
- Multiple linear regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$



- Polynomial regression

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$



Linear regression

Our goal is to find the best line that describes a linear relationship:

Find (β_0, β_1) where

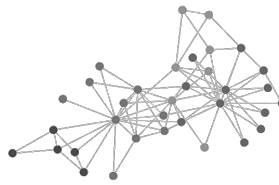
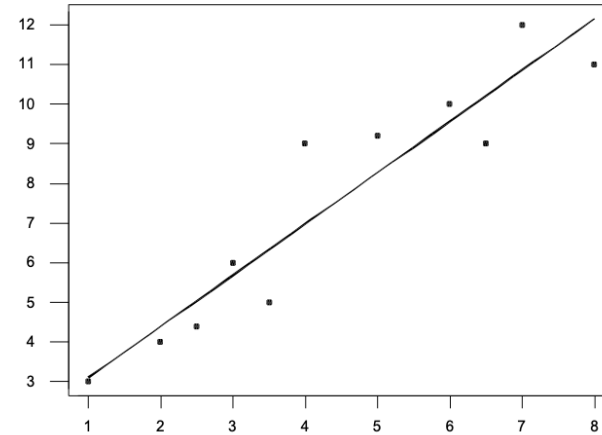
$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Unknown parameters:

1. β_0 *Intercept* (where the line crosses y-axis)
2. β_1 *Slope* of the line

Basic idea

- a. Plot observations (X, Y)
- b. Find best line that follows plotted points



Linear regression

With observed data $\{(X_1, Y_1), \dots, (X_n, Y_n)\}$, we model the linear relationship

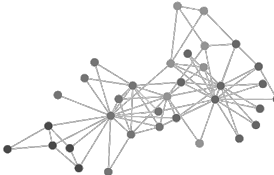
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, i = 1, \dots, n$$

$$E(\varepsilon_i) = 0$$

$$\text{Var}(\varepsilon_i) = \sigma^2$$

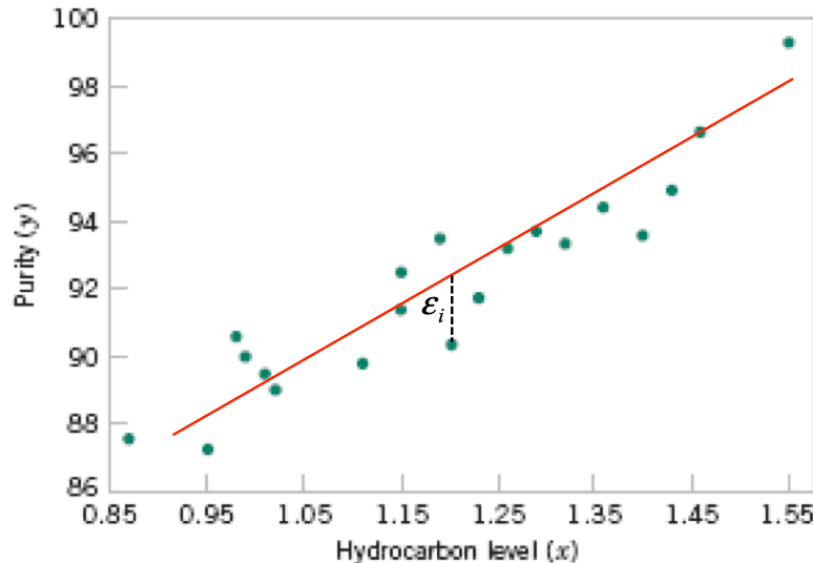
$\{\varepsilon_1, \dots, \varepsilon_n\}$ are independent random variables

(Later we assume $\varepsilon_i \sim \text{Normal}$)



Linear regression

Based on the scatter diagram, it is probably reasonable to assume that the mean of the random variable Y is related to X by the following **simple linear regression model**:



$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad i = 1, 2, \dots, n$$

$\epsilon_i \sim N(0, \sigma^2)$

Response

Regressor or Predictor

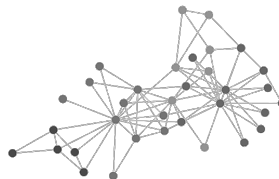
Intercept

Slope

Random error

where the slope and intercept of the line are called **regression coefficients**.

- The case of simple linear regression considers a single regressor or predictor x and a dependent or response variable Y.

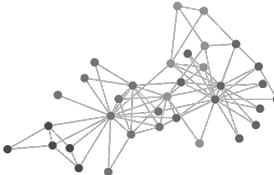


Estimate regression parameters

To estimate (β_0, β_1) , we find values that minimize squared error:

$$\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

- derivation: method of least squares



Method of least squares

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n$$

To estimate (β_0, β_1) , we find values that minimize squared error:

$$L = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

The least squares estimators of β_0 and β_1 , say, $\hat{\beta}_0$ and $\hat{\beta}_1$, must satisfy

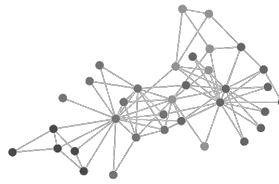
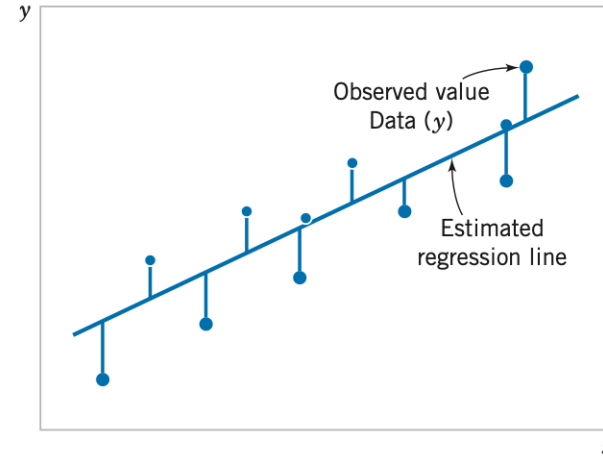
$$\left. \frac{\partial L}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\left. \frac{\partial L}{\partial \beta_1} \right|_{\hat{\beta}_0, \hat{\beta}_1} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

Least square normal equations

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

$$\hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n y_i x_i$$



Least square estimates

The **least squares estimates** of the intercept and slope in the simple linear regression model are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad (11-7)$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}} \quad (11-8)$$

where $\bar{y} = (1/n) \sum_{i=1}^n y_i$ and $\bar{x} = (1/n) \sum_{i=1}^n x_i$.



Alternative notation

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n} \quad (11-10)$$

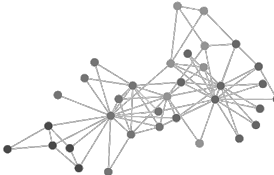
$$S_{xy} = \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^n x_i y_i - \frac{\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n} \quad (11-11)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Fitted (estimated)
regression model



Section two: Examples



Example: oxygen and hydrocarcon level

Table 11-1 Oxygen and Hydrocarbon Levels

Observation Number	Hydrocarbon Level x (%)	Purity y (%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

Question: fit a simple regression model to related purity (y) to hydrocarbon level (x)

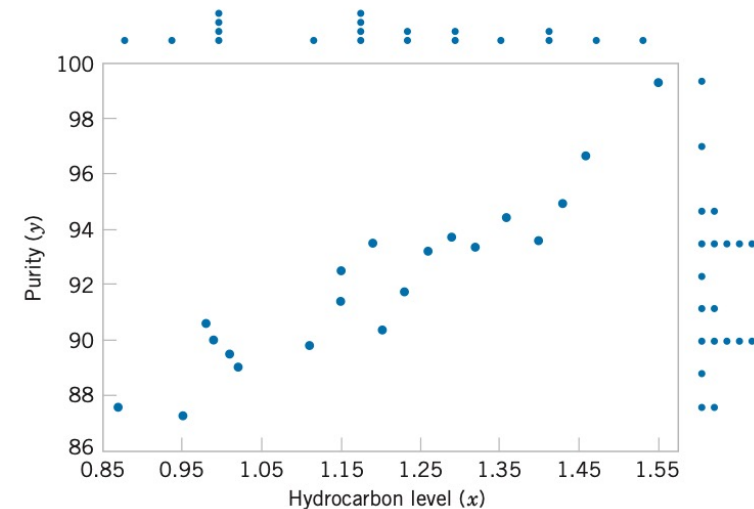
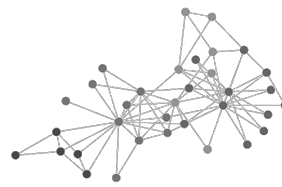


Figure 11-1 Scatter diagram of oxygen purity versus hydrocarbon level from Table 11-1.



Calculation 2

$$n = 20 \quad \sum_{i=1}^{20} x_i = 23.92 \quad \sum_{i=1}^{20} y_i = 1,843.21$$

$$\bar{x} = 1.1960 \quad \bar{y} = 92.1605$$

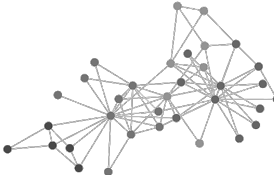
$$\sum_{i=1}^{20} y_i^2 = 170,044.5321 \quad \sum_{i=1}^{20} x_i^2 = 29.2892$$

$$\sum_{i=1}^{20} x_i y_i = 2,214.6566$$

$$\begin{aligned} S_{xx} &= \sum_{i=1}^{20} x_i^2 - \frac{\left(\sum_{i=1}^{20} x_i\right)^2}{20} = 29.2892 - \frac{(23.92)^2}{20} \\ &= 0.68088 \end{aligned}$$

and

$$\begin{aligned} S_{xy} &= \sum_{i=1}^{20} x_i y_i - \frac{\left(\sum_{i=1}^{20} x_i\right)\left(\sum_{i=1}^{20} y_i\right)}{20} \\ &= 2,214.6566 - \frac{(23.92)(1,843.21)}{20} = 10.17744 \end{aligned}$$



Calculation

Therefore, the least squares estimates of the slope and intercept are

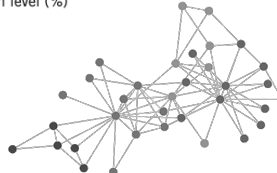
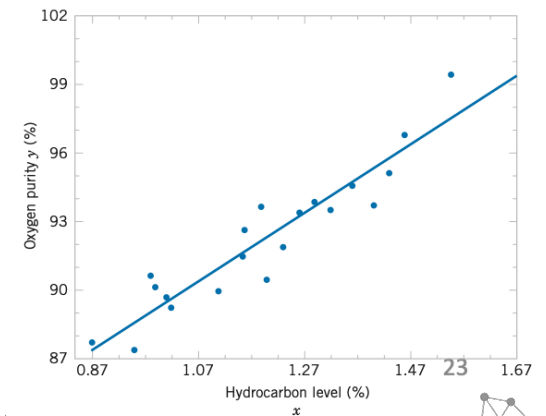
$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{10.17744}{0.68088} = 14.94748$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 92.1605 - (14.94748)1.196 = 74.28331$$

The fitted simple linear regression model (with the coefficients reported to three decimal places) is

$$\hat{y} = 74.283 + 14.947x$$



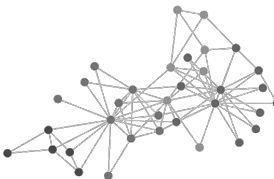
Interpretation of regression model

- Regression model

$$\hat{y} = 74.283 + 14.947x$$

$\hat{y} = 89.23\%$ when the hydrocarbon level is $x = 1.00\%$.

- This may be interpreted as an estimate of the true population **mean** purity when $x = 1.00\%$,
- The estimates are subject to error
- later: we will use confidence intervals to describe the error in estimation from a regression model



Estimation of variance

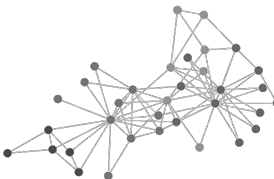
- Using the fitted model, we can estimate value of the response variable for given predictor

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- Residuals: $r_i = y_i - \hat{y}_i$
- Our model: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, i = 1, \dots, n, \text{Var}(\varepsilon_i) = \sigma^2$
- Unbiased estimator (MSE: Mean Square Error)

$$\hat{\sigma}^2 = \text{MSE} = \frac{\sum_{i=1}^n r_i^2}{n-2}$$

- oxygen and hydrocarcon level example $\hat{\sigma}^2 = 1.18$



Sammary

- Simple linear regression

$$Y = \beta_0 + \beta_1 X$$

- Estimate coefficients from data: method of least squares

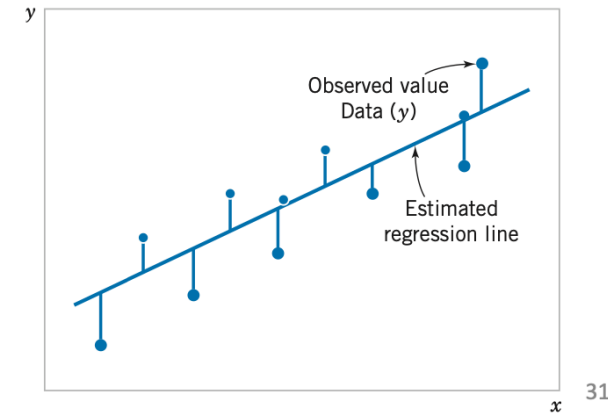
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Fitted (estimated)
regression model

- Estimate of variance



Example

- `import pandas as pd # for data manipulation`
`import numpy as np # for data manipulation`
`from sklearn.linear_model import LinearRegression # for creating a model`
`import plotly.graph_objects as go # for visualizations`
`import plotly.express as px # for visualizations`
- `# Read data into a Pandas DataFrame`
`df = pd.read_csv('Real estate.csv', encoding='utf-8')`

`# Print DataFrame`
`df`



Data

	No	X1 transaction date	X2 house age	X3 distance to the nearest MRT station	X4 number of convenience stores	X5 latitude	X6 longitude	Y house price of unit area
0	1	2012.917	32.0	84.87882	10	24.98298	121.54024	37.9
1	2	2012.917	19.5	306.59470	9	24.98034	121.53951	42.2
2	3	2013.583	13.3	561.98450	5	24.98746	121.54391	47.3
3	4	2013.500	13.3	561.98450	5	24.98746	121.54391	54.8
4	5	2012.833	5.0	390.56840	5	24.97937	121.54245	43.1
...	...	...	...	...	...	...	...	...
409	410	2013.000	13.7	4082.01500	0	24.94155	121.50381	15.4
410	411	2012.667	5.6	90.45606	9	24.97433	121.54310	50.0
411	412	2013.250	18.8	390.96960	7	24.97923	121.53986	40.6
412	413	2013.000	8.1	104.81010	5	24.96674	121.54067	52.5
413	414	2013.500	6.5	90.45606	9	24.97433	121.54310	63.9

414 rows × 8 columns

Joseph, B. (2020, June 17). *Linear Regression Made Easy: How Does It Work and How to Use It in Python*. Towards Data Science. <https://towardsdatascience.com/linear-regression-made-easy-how-does-it-work-and-how-to-use-it-in-python-be0799d2f159>



Code 1

- **# Create a scatter plot**
fig = px.scatter(df, x=df['X3 distance to the nearest MRT station'], y=df['Y house price of unit area'],
opacity=0.8, color_discrete_sequence=['black'])

Change chart background color
fig.update_layout(dict(plot_bgcolor = 'white'))

Update axes lines
fig.update_xaxes(showgrid=True, gridwidth=1, gridcolor='lightgrey',
zeroline=True, zerolinewidth=1, zerolinecolor='lightgrey',
showline=True, linewidth=1, linecolor='black')

fig.update_yaxes(showgrid=True, gridwidth=1, gridcolor='lightgrey',
zeroline=True, zerolinewidth=1, zerolinecolor='lightgrey',
showline=True, linewidth=1, linecolor='black')

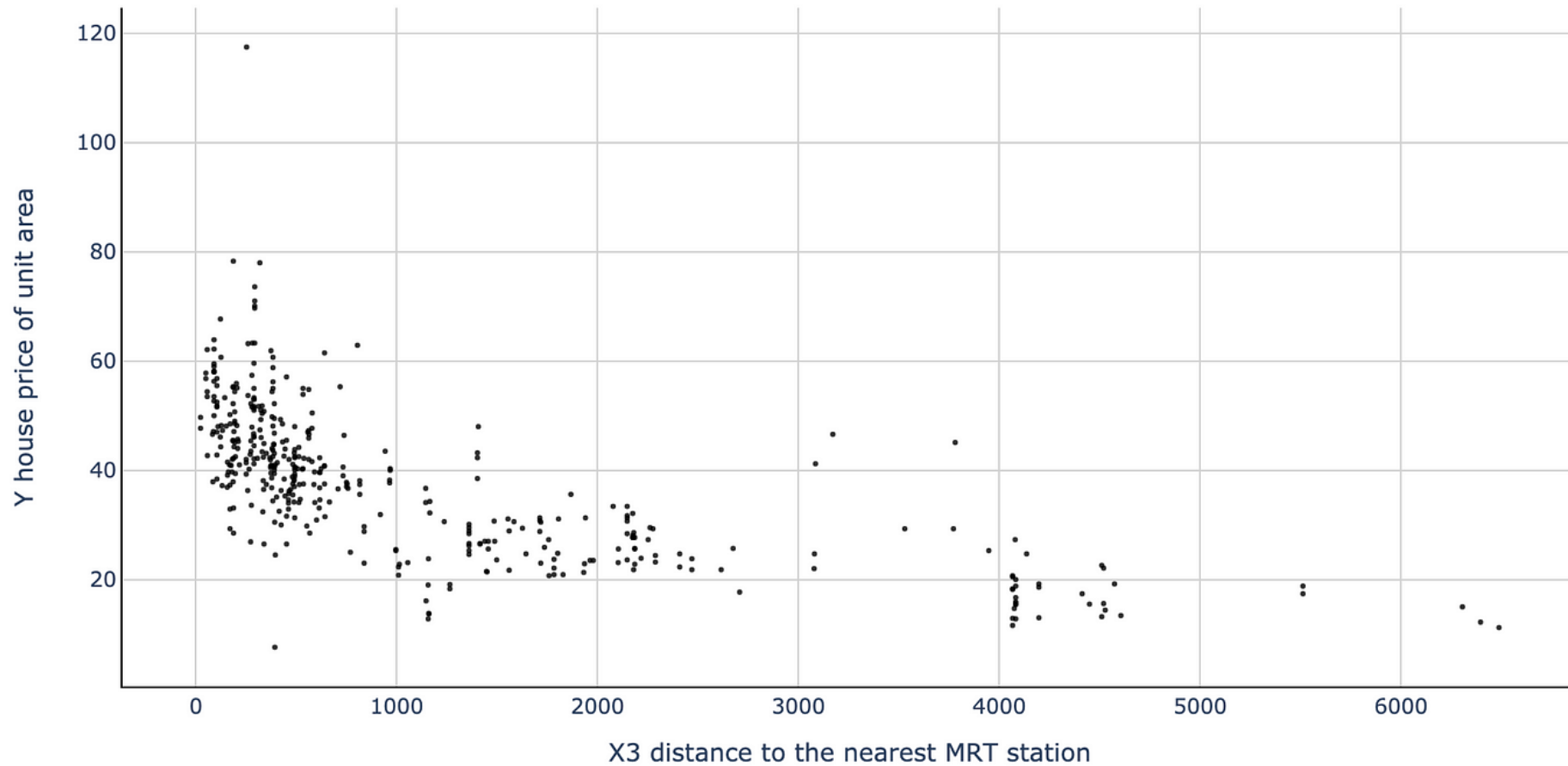
Set figure title
fig.update_layout(title_text="Scatter Plot")

Update marker size
fig.update_traces(marker=dict(size=3))

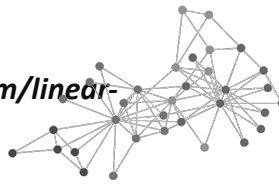
fig.show()



Scatter plot



Joseph, B. (2020, June 17). *Linear Regression Made Easy: How Does It Work and How to Use It in Python*. Towards Data Science. <https://towardsdatascience.com/linear-regression-made-easy-how-does-it-work-and-how-to-use-it-in-python-be0799d2f159>



Training

- # Select variables that we want to use in a model
Note, we need X to be a 2D array, hence reshape
X=df['X3 distance to the nearest MRT station'].values.reshape(-1,1)
y=df['Y house price of unit area'].values

Fit linear regression model
model = LinearRegression()
reg = model.fit(X, y)

Print the slope and intercept of the best-fit line
print(reg.coef_)
print(reg.intercept_)



Code 2

```
# Create 20 evenly spaced points from smallest X to largest X
x_range = np.linspace(X.min(), X.max(), 20)

# Predict y values for our set of X values
y_range = model.predict(x_range.reshape(-1, 1))

# Create a scatter plot
fig = px.scatter(df, x=df['X3 distance to the nearest MRT station'], y=df['Y house price of unit area'],
opacity=0.8, color_discrete_sequence=['black'])

# Add a best-fit line
fig.add_traces(go.Scatter(x=x_range, y=y_range, name='Regression Fit'))

# Change chart background color
fig.update_layout(dict(plot_bgcolor = 'white'))

# Update axes lines
fig.update_xaxes(showgrid=True, gridwidth=1, gridcolor='lightgrey',
zeroline=True, zerolinewidth=1, zerolinecolor='lightgrey',
showline=True, linewidth=1, linecolor='black')

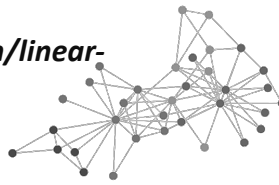
fig.update_yaxes(showgrid=True, gridwidth=1, gridcolor='lightgrey',
zeroline=True, zerolinewidth=1, zerolinecolor='lightgrey',
showline=True, linewidth=1, linecolor='black')

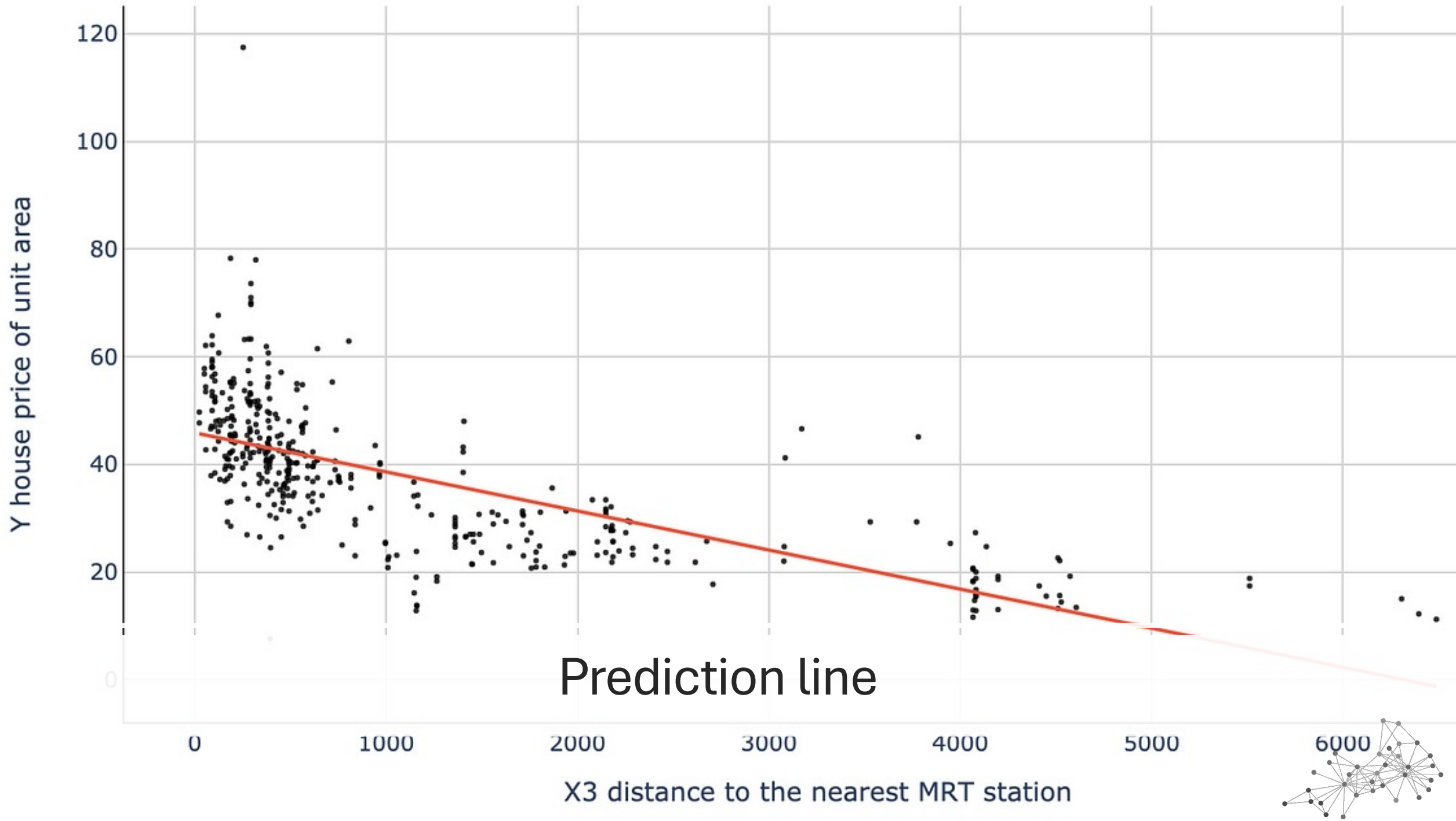
# Set figure title
fig.update_layout(title_text="Scatter Plot with Linear Regression Line")

# Update marker size
fig.update_traces(marker=dict(size=3))

fig.show()
```

Joseph, B. (2020, June 17). Linear Regression Made Easy: How Does It Work and How to Use It in Python. Towards Data Science. <https://towardsdatascience.com/linear-regression-made-easy-how-does-it-work-and-how-to-use-it-in-python-be0799d2f159>



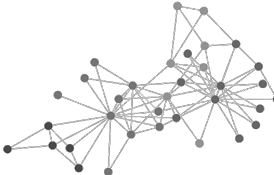


Multiple linear regression

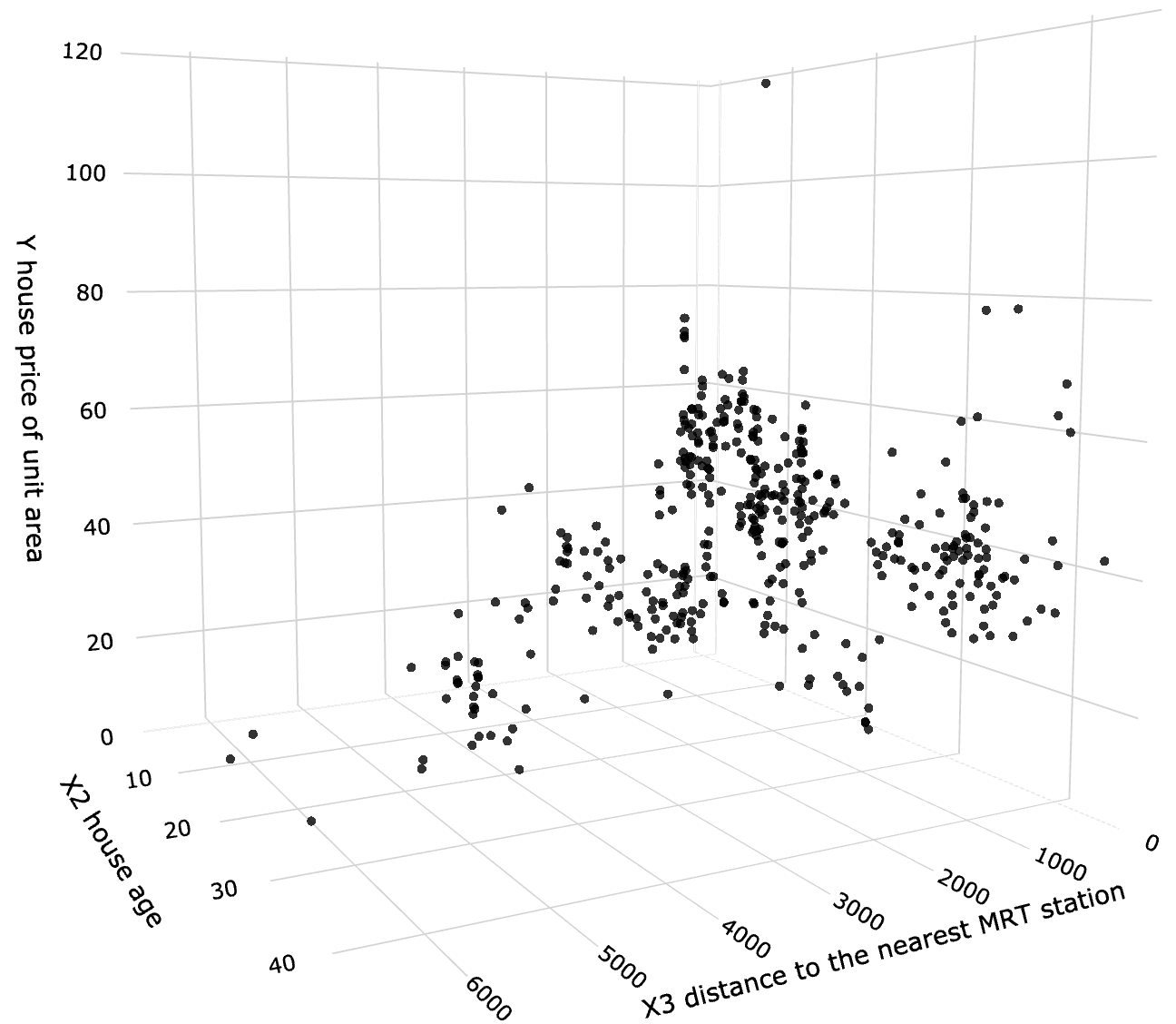
- `# Select variables that we want to use in a model`
`# Note, X in this case is already a 2D array, hence no reshape`
`X=df[['X3 distance to the nearest MRT station','X2 house age']]`
`y=df['Y house price of unit area'].values`

`# Fit linear regression model`
`model = LinearRegression()`
`reg = model.fit(X, y)`

`# Print slope(s) and intercept`
`print(reg.coef_)`
`print(reg.intercept_)`



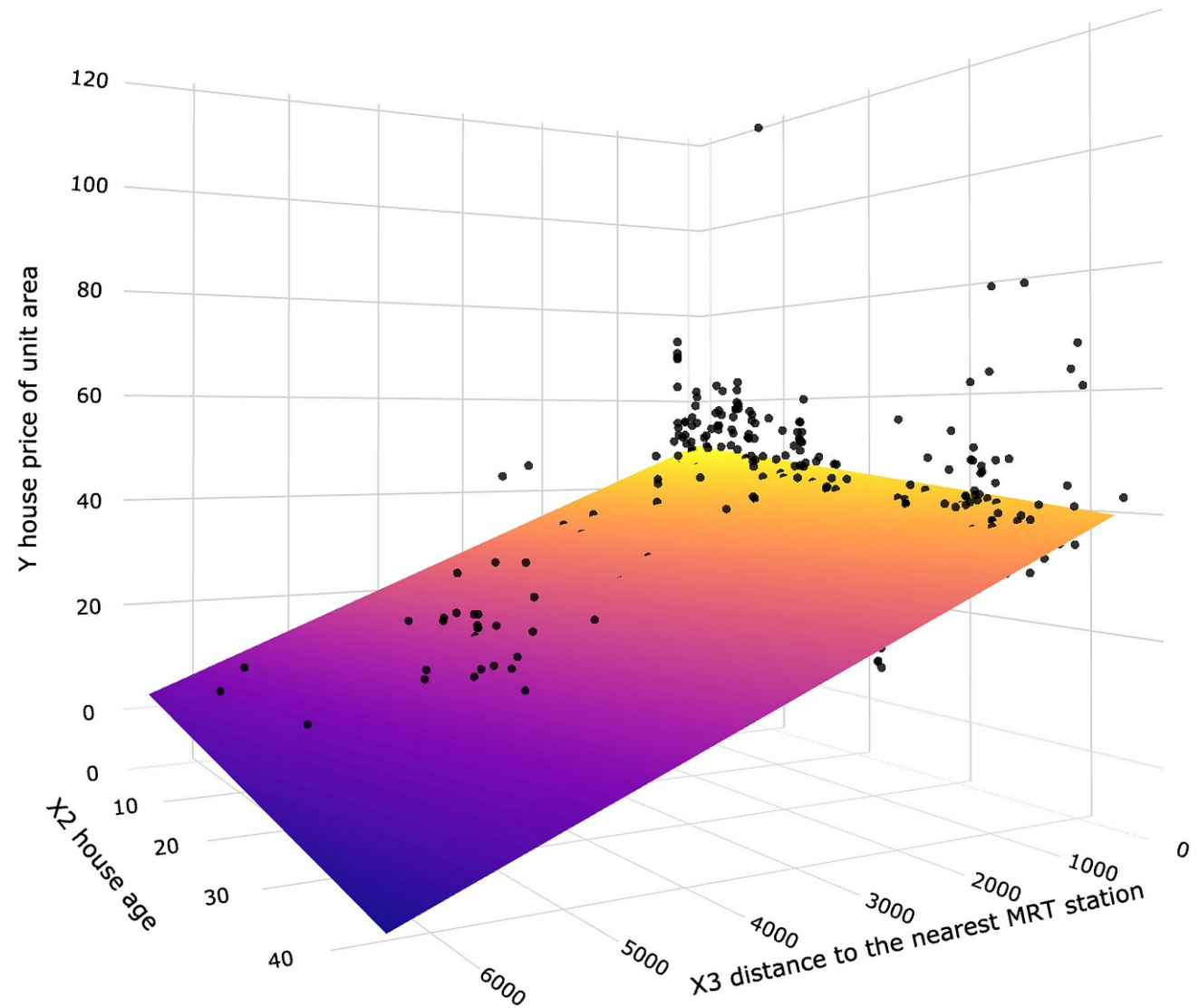
Multiple linear regression — Python example



Joseph, B. (2020, June 17). Linear Regression Made Easy: How Does It Work and How to Use It in Python. Towards Data Science. <https://towardsdatascience.com/linear-regression-made-easy-how-does-it-work-and-how-to-use-it-in-python-be0799d2f159>



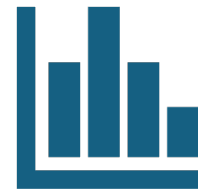
Fitted multiple linear regression



Joseph, B. (2020, June 17). Linear Regression Made Easy: How Does It Work and How to Use It in Python. Towards Data Science. <https://towardsdatascience.com/linear-regression-made-easy-how-does-it-work-and-how-to-use-it-in-python-be0799d2f159>



Section three: basic statistics



Basic statistics: Mean

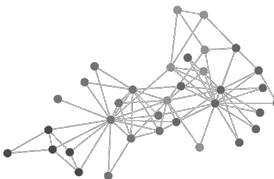
- The sample mean is the sum of all the observations ($\sum X_i$) divided by the number of observations (n):

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} \quad \text{where } \sum X_i = X_1 + X_2 + X_3 + X_4 + \dots + X_n$$

- **Example.** 1, 2, 2, 4, 5, 10. Calculate the mean. Note: n = 6 (six observations)

$$\sum X_i = 1 + 2 + 2 + 4 + 5 + 10 = 24$$

$$\bar{X} = 24 / 6 = \mathbf{4.0}$$



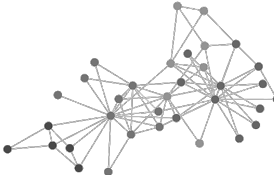
The median



The median is the middle value of the *ordered* data



To get the median, we must first rearrange the data into an ***ordered array*** (in ascending or descending order). Generally, we order the data from the lowest value to the highest value.



The mode

- The mode is the value of the data that occurs with the greatest frequency.

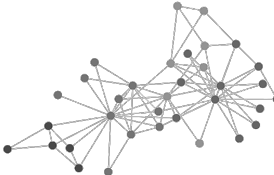
Example. 1, 1, 1, 2, 3, 4, 5

Answer. The mode is 1 since it occurs three times. The other values each appear only once in the data set.

Example. 5, 5, 5, 6, 8, 10, 10, 10.

Answer. The mode is: 5, 10.

There are two modes. This is a ***bi-modal*** dataset.

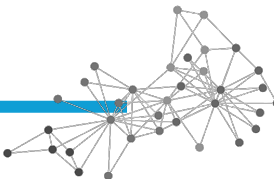


Standart deviation

- The standard deviation, s , measures a kind of “average” deviation about the mean. It is not really the “average” deviation, even though we may think of it that way.
- Why can't we simply compute the average deviation about the mean, if that's what we want?

$$\frac{\sum_{i=1}^n (X_i - \bar{X})}{n}$$

- If you take a simple mean, and then add up the deviations about the mean, as above, this sum will be equal to 0. Therefore, a measure of “average deviation” will not work.



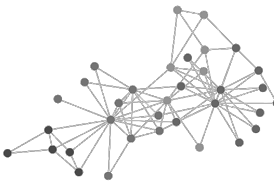
Standard Deviation

- Instead, we use:

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

- This is the “definitional formula” for standard deviation.
- The standard deviation has lots of nice properties, including:
 - By squaring the deviation, we eliminate the problem of the deviations summing to zero.
 - In addition, this sum is a minimum. No other value subtracted from X and squared will result in a smaller sum of the deviation squared. This is called the “least squares property.”
- Note we divide by (n-1), not n. This will be referred to as a loss of one degree of freedom.

Standard deviation is more intuitive than variance because it expresses the spread in the same units as the original data. It tells you how much, on average, data points deviate from the mean.



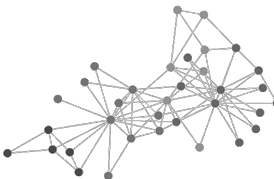
Variance

The variance, s^2 , is the standard deviation (s) squared. Conversely, $s = \sqrt{\text{variance}}$.

Definitional formula:
$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

Computational formula:
$$s^2 = \frac{\sum_{i=1}^n (X_i)^2 - \frac{(\sum_{i=1}^n X_i)^2}{n}}{n-1}$$

it quantifies how much the data points differ from the mean on average, squared.





Thank you for
your attention

